

HIMANSHU BADHANI

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RESEARCH INTERESTS

My primary interests are open quantum systems, with a focus on understanding thermodynamic and information-theoretic properties of noisy processes.

Topics that I am currently fascinated with:

- Stochastic thermodynamics, focusing on the fluctuation theorems for quantum processes.
- Non-Hermitian quantum mechanics, with a focus on the formalism of pseudo-Hermitian quantum mechanics.
- Quantum information theory, with a focus on information-theoretic properties of the quantum channels.
- Quantum thermodynamics of quantum channels and higher-order quantum processes.
- Algebraic quantum mechanics and the structure of C^* -algebras.
- Geometric features in quantum mechanics, specifically formalizing thermodynamics in a “curved” Hilbert space.

Topics that I plan to explore in the near future:

- Understanding the dynamical causal structure in many-body systems.
- Quantum simulations using open quantum system techniques.
- Topics in quantum sensing, e.g., using non-Hermiticity-enhanced sensing in NV centers for biological systems.

EMPLOYMENT

Postdoctoral fellow at the Center for Quantum Science and Technology, IIIT Hyderabad	Feb 2025–Present
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EDUCATION

Ph.D. The Institute of Mathematical Sciences, Chennai, India. Thesis: <i>Subsystem dynamics in pseudo-Hermitian quantum mechanics</i> . Submitted on November 27, 2024. Defended on August 04, 2025. Advisor: Dr. C M Chandrashekar.	2018-2025
BS-MS Indian Institute of Science Education and Research (IISER), Pune, India. Thesis: <i>Quantum fields in de-Sitter space</i> Advisor: Dr. Alok Laddha, CMI Chennai	2011 - 2016 CGPA: 8.2/10

Gap year: In 2017-2018 I took a gap year and explored small scale farming in my village. I am passionate about reforestation efforts in the Himalayas and am experimenting with low intervention farming. I also got interested in the mathematical theory of finance and briefly joined the doctoral program in finance at IIM Kolkata, a premier management institute in India, in 2018.

RESEARCH VISITS

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| 1. Research internship
Harish-Chandra Research Institute, Allahabad, India.
Project title: <i>de-Sitter stability and the inflationary potential</i>
Advisor: Dr. Dileep Jatkar | 2016–2017 |
| 2. Visiting researcher, IIIT Hyderabad, | Jan. 2024. |
| 3. Visiting postdoc, IMSc Chennai, | Nov. 2024-Jan. 2025, and May 2026. |

TEACHING EXPERIENCE

- I am currently mentoring two PhD students at IIIT Hyderabad
 - [1] Shradha Nair, 2nd year. Project title: Finding signatures of chaos in the conditional entropy of channels.
 - [2] Dhanuja G S, 4th year. Project title: Role of signaling in thermodynamics.
- Tutoring:
 - * Quantum Information Theory by Dr. Siddhartha Das, Monsoon 2025
 - * Introduction to Quantum Information by Dr. C. M. Chandrashekar, Spring 2020
- Mentored a few undergraduate summer students at IMSc over the years.

PUBLICATIONS

[Google scholar](#) / [ArXiv](#)

1. **HB** and S. Das, Conditional channel entropy sets fundamental limits on thermodynamic quantum information processing, arXiv:2604.01217 [quant-ph] (2026)
2. **HB**, Dhanuja G. S, and S. Das, Thermodynamic work capacity of quantum information processing, [APS Open Sci.](#) **1**, 000092, (2026) [[Editor's suggestion](#)].
3. **HB**, Dhanuja G. S, and S. Das, Thermodynamics of quantum processes: An operational framework for free energy and reversible athermalities, arXiv:2510.12790 [quant-ph] (2025).
4. **HB**, Dhanuja G. S., S. Choudhary, V. Anand, and S. Das. Erasure cost of a quantum process: a thermodynamic meaning of the dynamical min-entropy, [Quantum Sci. Technol.](#) **11** 015038, (2026).
5. **HB**, C. M. Chandrashekar. Reduced dynamics in quasi-Hermitian systems, [Annals of Physics](#), Volume 484, 170304, (2026).
6. R. Basu, A. Chakraborty, **HB**, M. Alimuddin, and S. Bhattacharya. Exploring non-markovianity in ergodic channels: Measuring memory retention through ergotropy. [Phys. Rev. A](#) **111**, 032416, (2025).
7. F. Buscemi, R. Gangwar, K. Goswami, **HB**, T. Pandit, B. Mohan, S. Das, and M. N. Bera. Causal and noncausal revivals of information: A new regime of non-markovianity in quantum stochastic processes. [PRX Quantum](#) **6**, 020316, (2025).
8. **HB** and S. Ghosh. Decomposition of a system in pseudo-hermitian quantum mechanics. [J. Phys. A: Math. Theor.](#) **58** 165301, (2025).
9. **HB**, S. Banerjee, and C. M. Chandrashekar. Non-Hermitian quantum walks and non-Markovianity: the composition interaction. [Phys. Scr.](#) **99** 105112, (2024).
10. A. Mitra, **HB**, and S. Ghosh. Improvement in quantum communication using quantum switch. [Phys. Scr.](#) **98** 045101, (2023).
11. **HB** and C. M. Chandrashekar. Gravitationally induced entanglement dynamics between two quantum walkers. [Eur. Phys. J. C](#) **81**, 454, (2021).

CONTRIBUTED TALKS AND SEMINARS

1. (Upcoming) [21st CEQIP workshop](#), Mikulov, Czech Republic, Sept. 2026. **Title:** Thermodynamics of quantum information processing. (sponsored by [ANRF-ITS](#))
2. Physics seminar (online) at Heriot-Watt University, Edinburgh, June 2026. **Title:** Thermodynamic work capacity of quantum information processing.
3. Physics seminar at IMSc Chennai, May 2026, **Title:** Thermodynamic aspects of quantum processes.
4. [Physics seminar](#) at TIFR Hyderabad, May 2026, **Title:** Thermodynamic aspects of quantum processes.
5. Physics seminar at IIIT Hyderabad, Jan. 2024, **Title:** Virtual subsystems of non-trivial inner product spaces.
6. [Chennai Strings Meeting 2023](#), IMSc, Chennai, **Title:** Partial tracing in a non-trivial metric Hilbert space.
7. [PHHQP20 2023](#), Koc University, Istanbul, **Title:** PT-symmetric quantum walk: exploring the reduced dynamics.

8. [PiQUST 2023](#) IIT Madras, **Title:** Non-Hermitian quantum walks.
9. [IMSc60 2023](#), IMSc Chennai, **Title:** Non-Hermitian quantum walks.

POSTER PRESENTATIONS IN INTERNATIONAL CONFERENCES

1. [TQC 2025](#) IISc Bangalore, **Title:** Thermodynamic interpretation of dynamical min-entropy.
2. [QCMC 2024](#), IIT Madras, **Title:** Defining subsystems in Hilbert spaces with non-Euclidean metric.
3. [PQIQC 2024](#), S. N. Bose NCBS, Kolkata, **Title:** Virtual sub-systems in a non-trivial metric Hilbert space.
4. [AQIS 2021](#), University of Tokyo, Japan (online), **Title:** Non-Hermitian quantum walks: position-coin interaction.
5. [Indo-German Workshop](#) on Quantum Science and Technologies 2020, IIT Madras, **Title:** Gravitationally induced entanglement between two quantum walkers.
6. [QFTA 2019](#), IISER Mohali, **Title:** Gravitationally induced entanglement between two quantum walkers.

NUMERICAL SKILLS

- Languages: MATLAB, Mathematica, and Python.
- Numerical skills: Monte Carlo optimization, classical simulated annealing, semidefinite programming.
- QI tools: Qiskit, CVXPY, PennyLane.

PEER REVIEW AND OUTREACH

- **Peer review:** *Class. Quantum Grav.*, *J. Phys. A: Math. Theor.*, *Physica Scripta*, *Quantum Inf Process*, TQC 2024.
- Panel member in the workshop on [Impact of Quantum Technologies on Life Sciences](#), organized with [NITI Aayog](#), Gov. of India.
- YouTube manager for [Quantum Meets](#).
- Among the early members of [Disha](#) IISER Pune, a volunteer organization working on educational equity. I also started and ran a series of talks called *Talk for Twenty* and a magazine called *Pahal*, both of which are still going strong at IISER Pune.

LANGUAGES

Fluent: English, Hindi, Kumaoni
 Beginner: Tamil, Marathi, German (A1 level)

ACADEMIC MERITS AND AWARDS

- [Visvesvaraya Post-Doctoral Fellowship](#) (2025-2026) by the Ministry of Electronics and Information Technology (MeitY), Gov. of India.
- Rank 22 in the **Joint Entrance Screening Test (JEST) 2018** for PhD positions in India.
- Qualified for the **Junior Research Fellowship** by CSIR, Gov. of India in 2016. The qualification guarantees a 5-year fellowship for pursuing a PhD anywhere in India.
- *Excellent* grade for the MS thesis titled *Quantum fields in de-Sitter space*.
- INSPIRE fellowship (2011-2016) by the Dept. Of Science and Technology, Gov. of India

REFERENCES

Dr. C M Chandrashekar [✉](#)
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 Bangalore
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Dr. Sibasish Ghosh [✉](#)
 Institute of Mathematical Sciences,
 Chennai
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 Technology, Hyderabad
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